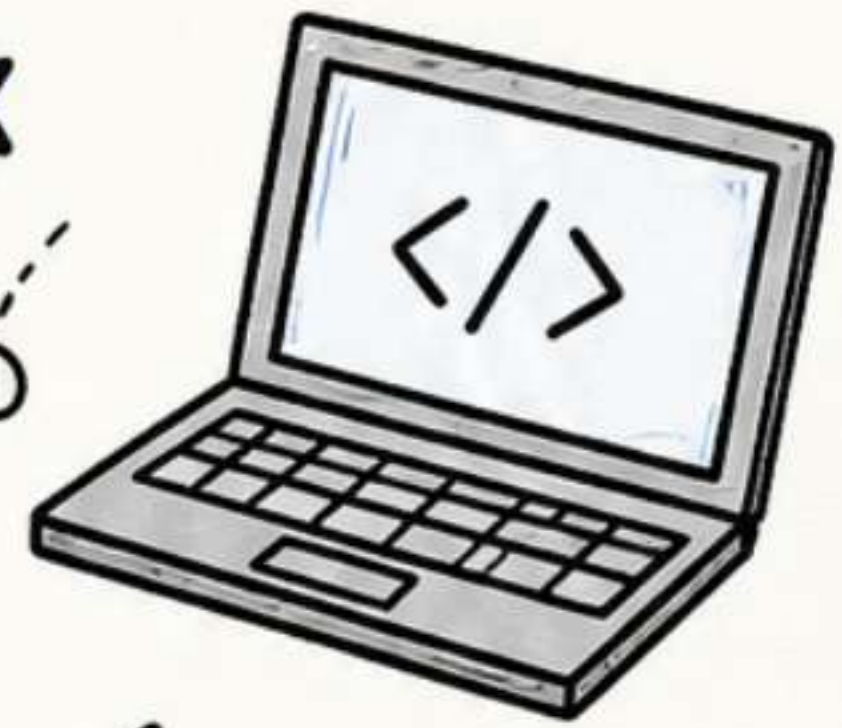


Learn
Practice
Crack!



ALL IN ONE

JAVA



+

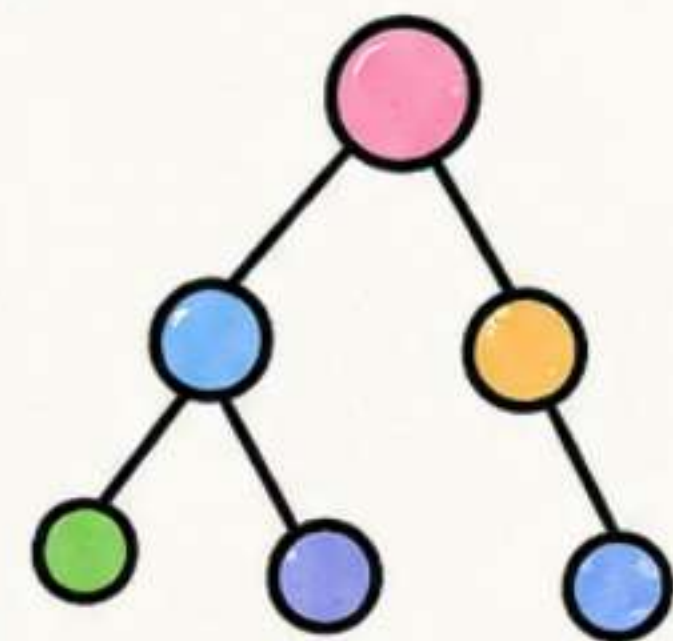
OOPS

Think
Code
Succeed



+

DSA



THEORY NOTES

for
IMPORTANT
INTERVIEW QUESTIONS



Good Notes
Better Understanding
Bright Future

KEY TO
SUCCESS:

- ☒ Consistent
- ☒ Practice
- ☒ Confidence
- ☒ Never Give Up





INDEX



JAVA



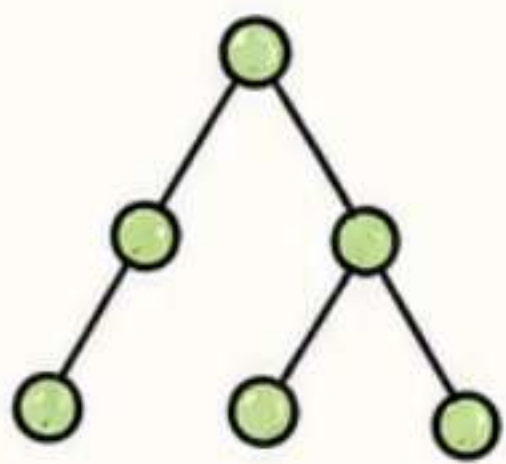
1. Java Basics
2. Operators & Expressions
3. Control Statements
4. Arrays
5. Strings
6. Exception Handling
7. Collection Framework
8. File Handling
9. Multithreading
10. Generics

OOPS CONCEPTS

1. Class & Object
2. Inheritance
3. Polymorphism
4. Abstraction
5. Encapsulation
6. Constructor
7. This Keyword
8. Super Keyword
9. Static Keyword
10. Final Keyword



DSA (DATA STRUCTURES & ALGORITHMS)



1. Time & Space Complexity
2. Arrays
3. Strings
4. Linked List
5. Stack
6. Queue
7. Trees
8. Binary Search Tree
9. Heaps / Priority Queue
10. Hashing

11. Graphs
12. Greedy Algorithms
13. Dynamic Programming
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15. Bit Manipulation
16. Important DSA Interview Questions
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18. DSA Cheat Sheet
19. Frequently Asked Java Interview Questions
20. Conclusion



Learn. Practice. Crack.
You've Got This!





JAVA INTERVIEW QUESTIONS (5)



1. Q: What is Java?

Ans: Java is a high-level, class-based, object-oriented programming language developed by Sun Microsystems.

2. Q: What are the main features of Java?

Ans: Simple, Object-Oriented, Platform Independent, Secure, Robust, Portable, Multithreaded, Architecture Neutral, High Performance, Distributed.



3. Q: What is JVM?

Ans: JVM (Java Virtual Machine) is an abstract machine that enables a computer to run a Java program. It converts bytecode into machine code and provides platform independence.

4. Q: What is the use of the main() method?

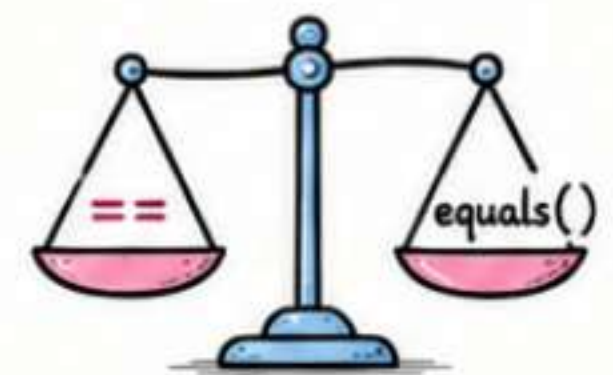
Ans: The main() method is the entry point of any Java program. Execution starts from the main() method.



5. Q: What is the difference between == and equals() in Java?

Ans:

- == compares reference (memory address).
- equals() compares content/value.





JAVA INTERVIEW QUESTIONS (5)



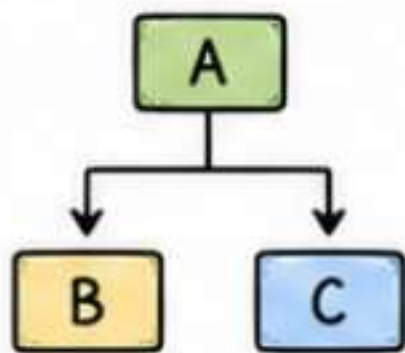
6. Q: What is the difference between String, StringBuilder and StringBuffer?

Ans: String is immutable (cannot be changed once created).
StringBuilder is mutable and not synchronized (faster).
StringBuffer is mutable and synchronized (thread-safe).



7. Q: What is the difference between primitive data types and reference data types?

Ans: Primitive data types store actual values (int, char, etc.) and are stored in stack. Reference data types store reference (address) to objects (arrays, classes, etc.) and are stored in heap.



8. Q: What is inheritance in Java?

Ans: Inheritance is a mechanism in Java where one class acquires the properties (fields and methods) of another class. It promotes code reusability.



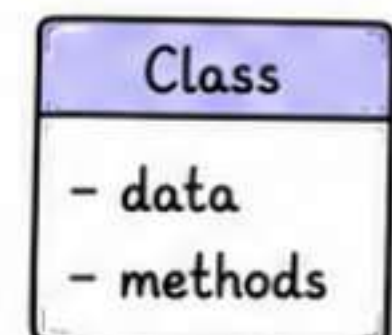
9. Q: What is polymorphism in Java?

Ans: Polymorphism means "many forms". It allows a method to do different things based on the object that it is acting upon. There are two types: Compile-time (method overloading) and Run-time (method overriding).



10. Q: What is encapsulation in Java?

Ans: Encapsulation is the process of wrapping data (variables) and code (methods) together as a single unit (class) and restricting direct access to some of the object's components.





JAVA INTERVIEW QUESTIONS (5)



● Q: What is the difference between Array and ArrayList in Java?

Ans: Array is a fixed-size data structure that can store elements of the same type. ArrayList is a resizable array implementation in the Collections framework that can grow or shrink dynamically.



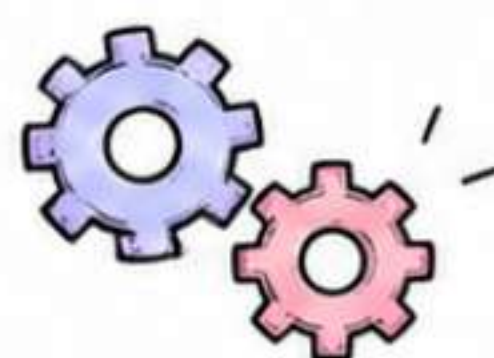
● Q: What is String in Java? Is String mutable or immutable?

Ans: String is a sequence of characters. In Java, String is immutable, which means once a String object is created, it cannot be changed.



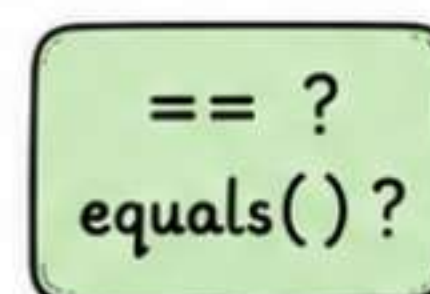
● Q: What is Exception Handling in Java?

Ans: Exception Handling is a mechanism to handle runtime errors (exceptions) so that normal flow of the application can be maintained. It is done using try, catch, finally, throw and throws.



● Q: What is the difference between == and equals() method?

Ans: == compares reference (memory address) of objects. equals() compares the content/value of objects.



● Q: What is the use of "this" keyword in Java?

Ans: "this" keyword refers to the current object. It is used to resolve ambiguity between instance variables and local variables, and to call current class methods or constructors.



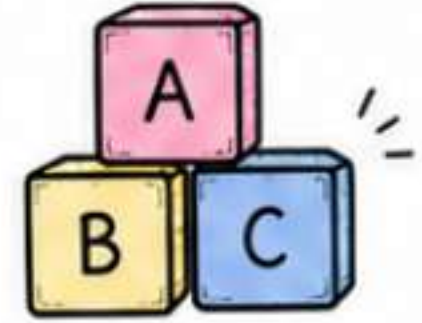
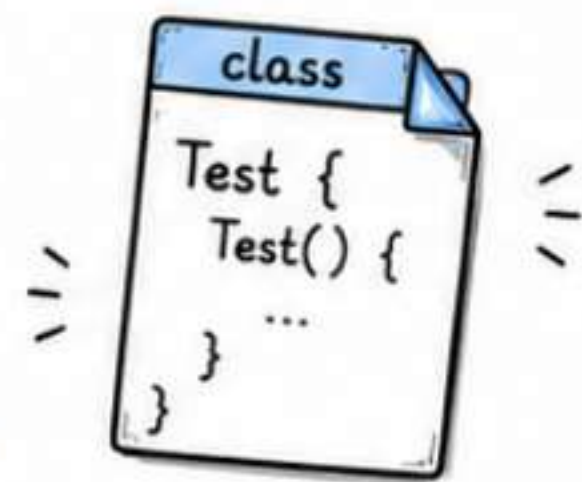


JAVA INTERVIEW QUESTIONS (5)



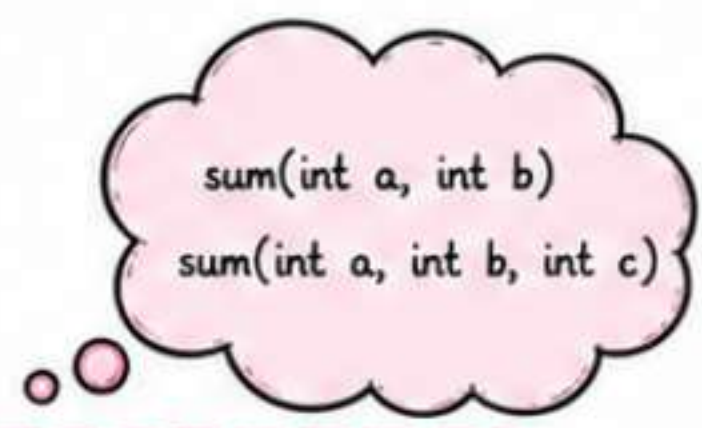
● What is a Constructor in Java?

Ans: A constructor is a special method used to initialize objects. It has the same name as the class and does not have a return type. It is called automatically when an object is created.



● What is Method Overloading in Java?

Ans: Method Overloading is a feature that allows a class to have multiple methods with the same name but different parameters (number, type or order of parameters).



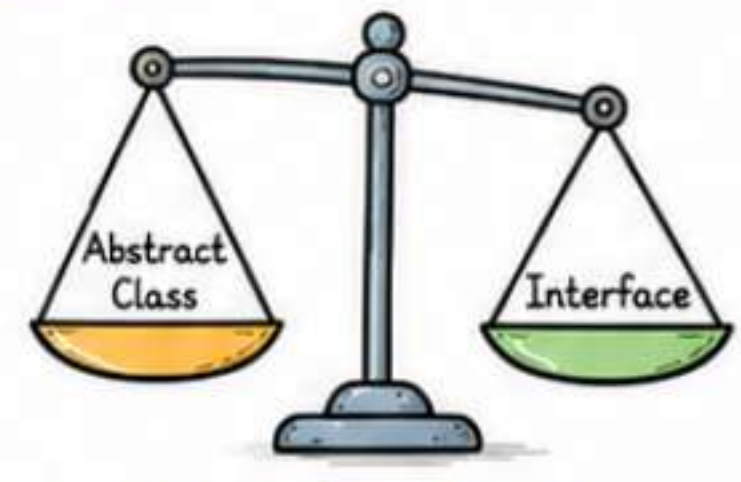
● What is the difference between final, finally and finalize?

Ans: final is a keyword used for variables, methods and classes (cannot be modified). finally is a block used with try-catch. It always executes, whether exception occurs or not. finalize() is a method of Object class called by Garbage Collector before destroying an object.



● What is the difference between interface and abstract class?

Ans: Abstract class can have abstract and concrete methods, instance variables, constructors and any access modifier. Interface can have only abstract methods (before Java 8), public static final variables. It cannot have constructors.



● What is the difference between checked and unchecked exceptions?

Ans: Checked Exceptions are checked at compile-time (e.g., IOException, SQLException). They must be handled using try-catch or throws. Unchecked Exceptions are checked at runtime (e.g., NullPointerException, ArithmeticException). They are subclasses of RuntimeException.



Practice Today, Perform Tomorrow!

IMPORTANT JAVA INTERVIEW QUESTIONS

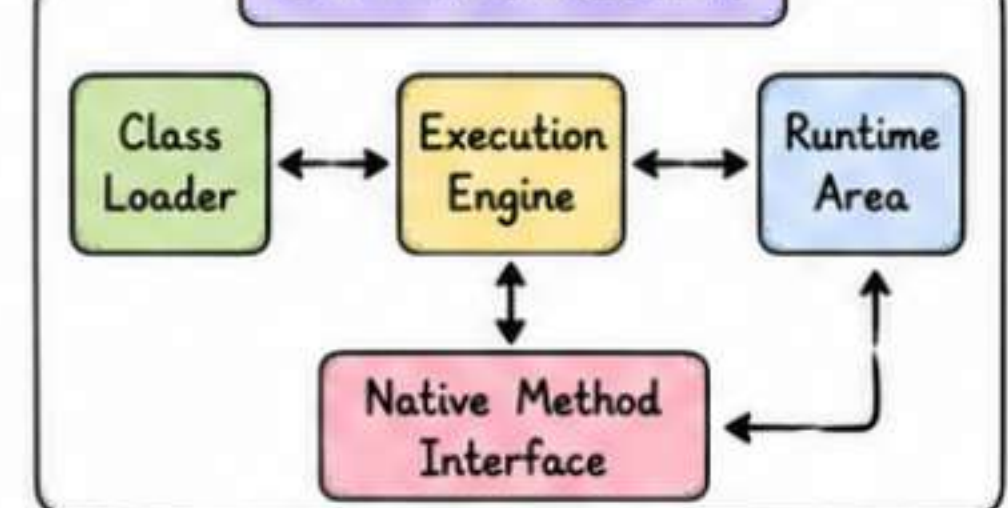


JVM

What is JVM?

Ans: JVM (Java Virtual Machine) is an abstract machine that allows a computer to run a Java program. It converts bytecode into machine code and provides platform independence.

JVM Architecture



.class

What is bytecode?

Ans: Bytecode is an intermediate code generated by the Java compiler (.class file) which is platform independent and executed by the JVM.

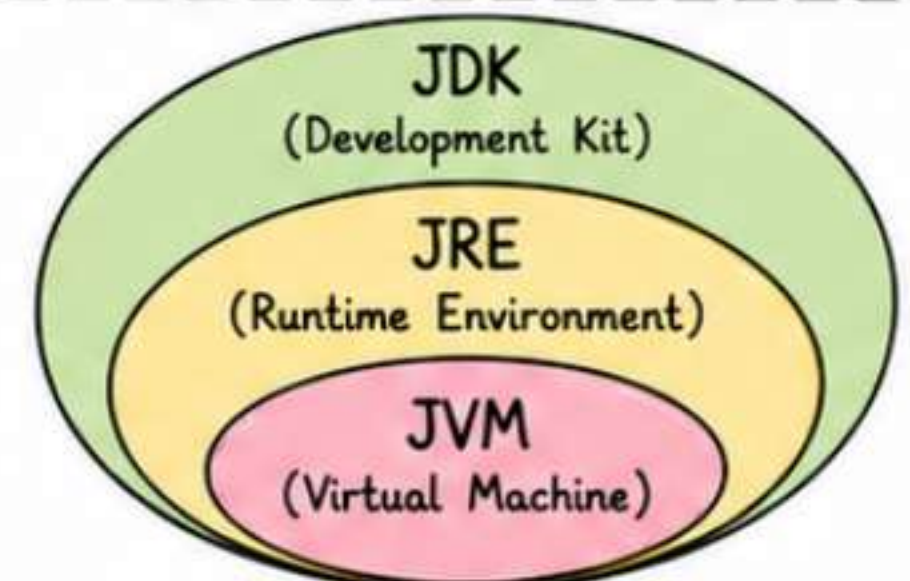


JDK vs JRE

What is the difference between JDK, JRE and JVM?

Ans:

- JDK (Java Development Kit): Provides tools for development (compiler, debugger, etc.).
- JRE (Java Runtime Environment): Provides libraries and JVM to run Java programs.
- JVM (Java Virtual Machine): Executes the bytecode.



main()

What is the use of main() method?

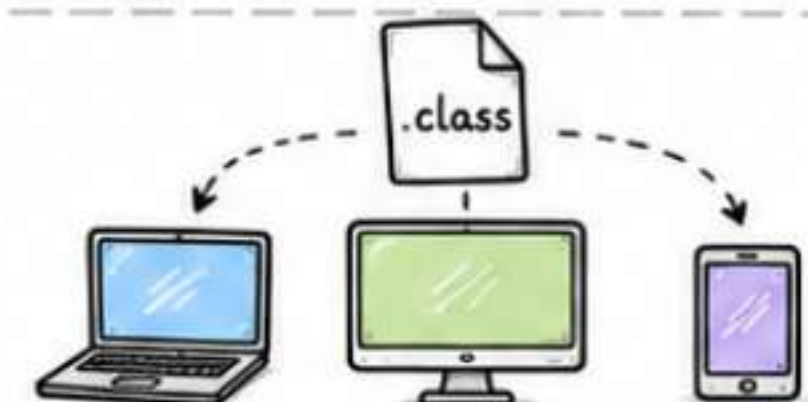
Ans: main() is the entry point of any Java program. Execution of a Java program starts from the main() method.

```
public class Test {  
    public static void main(String[] args) {  
        // code  
    }  
}
```

Platform Independence

What is platform independence in Java?

Ans: Java achieves platform independence because the same .class file (bytecode) can run on any platform that has a JVM. Write Once, Run Anywhere (WORA).



Prepare Well Succeed Always! 😊

IMPORTANT JAVA INTERVIEW QUESTIONS



Q: What is the difference between `==` and `equals()` in Java?

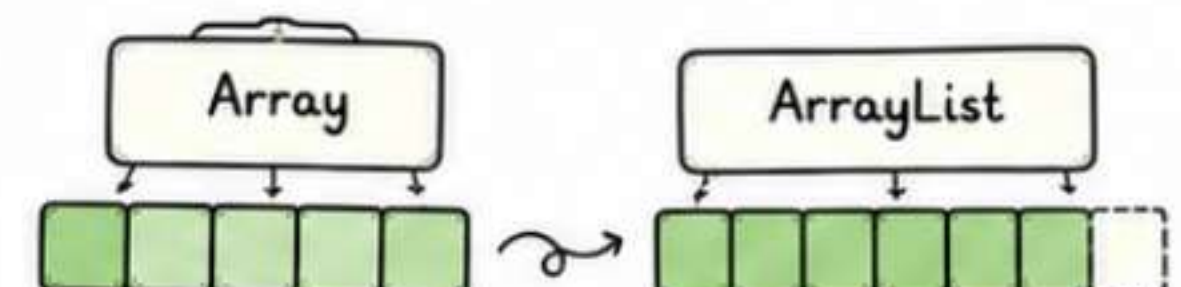
Ans: `==` compares reference (memory address) of objects.
`equals()` compares the content/value of objects.

<code>==</code>	<code>equals()</code>
Compare Reference	Compare Content



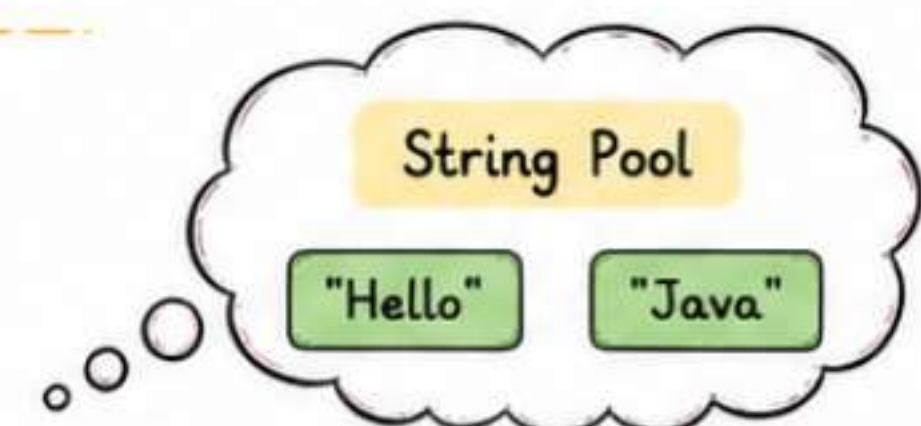
Q: What is the difference between Array and ArrayList?

Ans: Array is fixed-size. Once created, its size cannot be changed. ArrayList is resizable. It is part of Collections framework and can grow or shrink dynamically.



Q: What is String pool in Java?

Ans: String pool is a special memory area in heap where string literals (strings in double quotes) are stored. If a string literal already exists in the pool, same reference is returned.



Q: What is the difference between HashMap and Hashtable?

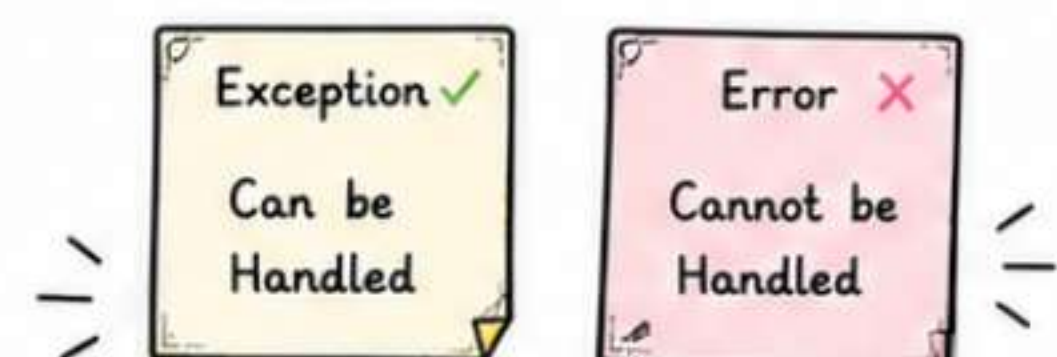
Ans: HashMap is not synchronized and not thread-safe. Allows null key and multiple null values. Hashtable is synchronized and thread-safe. Does not allow null key or null values.

Key	Value
K1	V1
K2	V2
...	...



Q: What is Exception and Error in Java?

Ans: Exception: Conditions that an application can catch and handle.
(e.g., IOException, NullPointerException)
Error: Serious problems that a program usually cannot handle.
(e.g., OutOfMemoryError, StackOverflowError)



Q: What is the purpose of static keyword in Java?

Ans: Static keyword is used for memory management. Static members (variables, methods, blocks) belong to the class, not to any object. They can be accessed using class name.



Prepare Well Succeed Always! 😊

IMPORTANT JAVA INTERVIEW QUESTIONS



Q: What is the difference between `==` and `equals()` in Java?

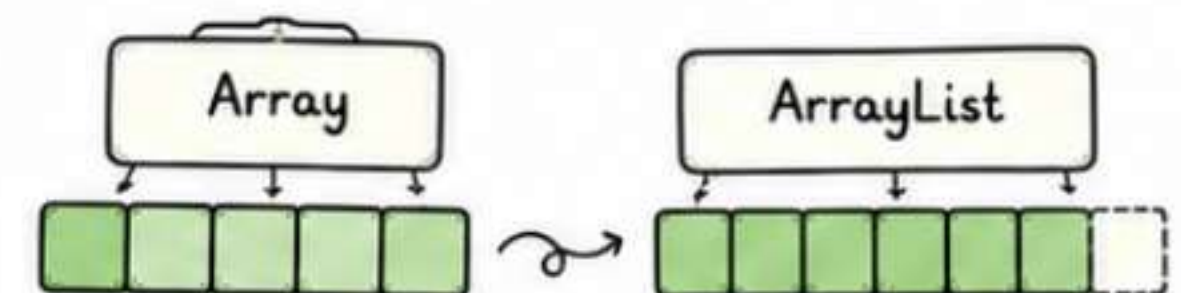
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<code>==</code>	<code>equals()</code>
Compare Reference	Compare Content



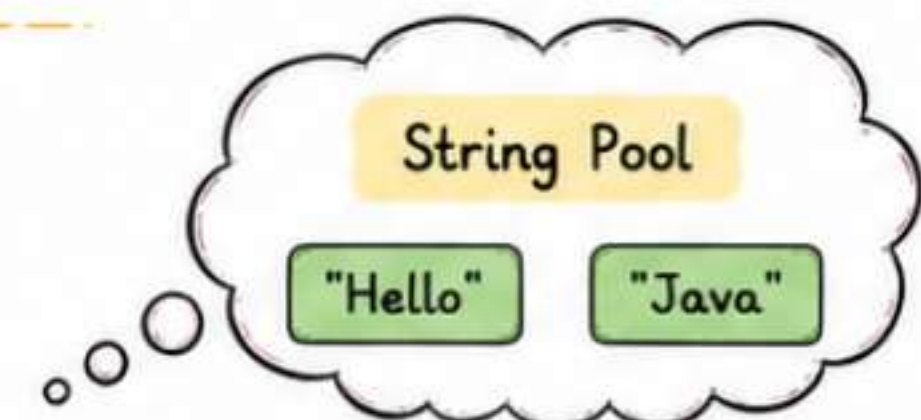
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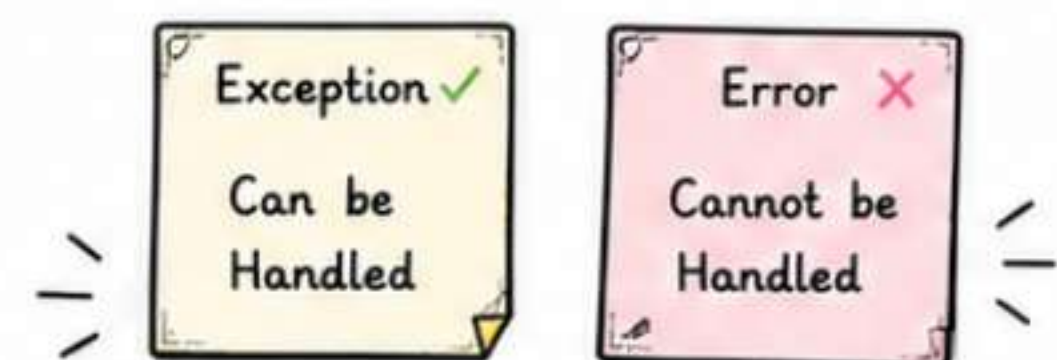
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Ans: Static keyword is used for memory management. Static members (variables, methods, blocks) belong to the class, not to any object. They can be accessed using class name.



Think OOPS,
Code Better!

OOPS INTERVIEW QUESTIONS



CLASS



● What is a Class in Java?

Ans: A class is a blueprint or template for creating objects.
It contains data (fields) and methods to operate on that data.



OBJECT

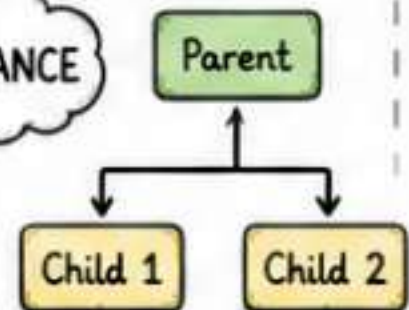


● What is an Object in Java?

Ans: An object is an instance of a class.
It has state (data) and behavior (methods).



INHERITANCE



● What is Inheritance in Java?

Ans: Inheritance is a mechanism where one class acquires the properties (fields and methods) of another class.
It promotes code reusability.



POLYMORPHISM



Same action, different results

● What is Polymorphism in Java?

Ans: Polymorphism allows objects to take many forms.
It is achieved by method overloading (compile-time) and method overriding (run-time).



ABSTRACTION



● What is Abstraction in Java?

Ans: Abstraction is a concept of hiding internal details and showing only essential features. It is achieved using abstract classes and interfaces.



OOPS Concepts

- ✓ Class
- ✓ Object
- ✓ Inheritance
- ✓ Polymorphism
- ✓ Abstraction
- ✓ Encapsulation

Master OOPS,
Master Java!

Practice OOPS,
Crack Interviews!

OOPS INTERVIEW QUESTIONS

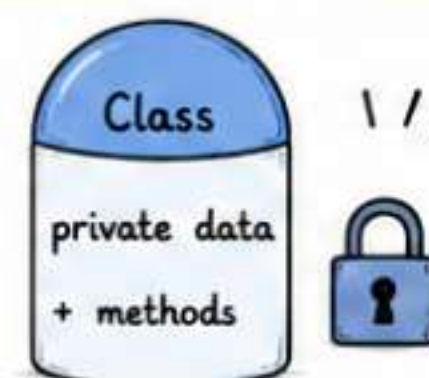


ENCAPSULATION



What is Encapsulation in Java?

Ans: Encapsulation is the wrapping of data (variables) and code (methods) together as a single unit. It restricts direct access to some of an object's components and can prevent accidental modification of data. It is achieved using private keywords, getters and setters.



CONSTRUCTOR



What is a Constructor in Java?

Ans: A constructor is a special method used to initialize objects. It has the same name as the class and has no return type. It is called automatically when an object is created.

```
class Student {  
    Student() {  
        // constructor  
    }  
}
```

THIS KEYWORD



What is 'this' keyword in Java?

Ans: 'this' keyword refers to the current object. It is used to resolve the ambiguity between instance variables and parameters, and to call current class methods/constructors.

```
class Student {  
    int id;  
    Student(int id) {  
        this.id = id; // refers to instance variable  
    }  
}
```

SUPER KEYWORD



What is 'super' keyword in Java?

Ans: 'super' keyword is used to refer immediate parent class object. It is used to access parent class variables, methods and constructors from the child class.

```
class Child extends Parent {  
    Child() {  
        super(); // calls parent constructor  
    }  
}
```

STATIC KEYWORD



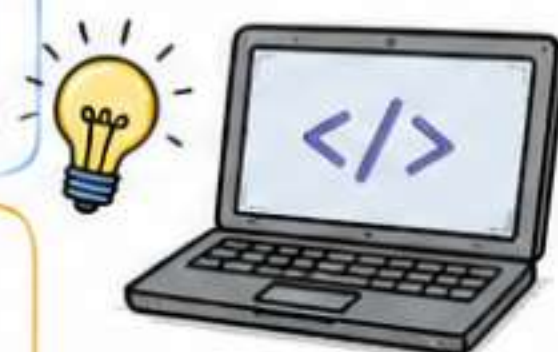
What is 'static' keyword in Java?

Ans: 'static' keyword is used for memory management. Static variables belong to the class, not to any object. Static methods can be called without creating an object.

```
class Demo {  
    static int count = 0; // class variable  
    static void show() { // static method  
        System.out.println("Hello");  
    }  
}
```

Remember:

- ✓ Classes create objects.
- ✓ Objects have state and behavior.
- ✓ OOP makes programs modular, reusable and easy to maintain.



Prepare Well,
Code Better!

MORE IMPORTANT OOPS INTERVIEW QUESTIONS



ABSTRACTION



What is Abstraction in OOPs?

Ans: Abstraction is the concept of hiding implementation details and showing only essential features to the user. It is achieved using abstract classes and interfaces.

```
abstract class Shape {  
    abstract void draw();  
}
```

Show What,
Not How!

INTERFACE



What is an Interface in Java?

Ans: An interface is a blueprint of a class. It contains only abstract methods (by default public & abstract) and constants (by default public, static & final).

```
interface Drawable {  
    void draw();  
}
```

Interface Points

- ✓ 100% Abstraction
- ✓ Multiple inheritance achieved using interfaces

DIFFERENCE: ABSTRACTION & ENCAPSULATION

What is the difference between Abstraction and Encapsulation?

Ans: Abstraction focuses on hiding what an object does. Encapsulation focuses on hiding the data (state) of an object and protecting it from outside access.

Abstraction
Hides
Implementation
(shows what)

+

Encapsulation
Hides Data
(protects how)



COMPOSITION



What is Composition in OOPs?

Ans: Composition is a strong form of association. In composition, one class contains an object of another class and controls its lifecycle.

```
class Engine { }  
class Car {  
    Engine engine = new Engine();  
}
```

Part-Whole
Relationship

AGGREGATION



What is Aggregation in OOPs?

Ans: Aggregation is a weak form of association. The child object can exist independently of the parent object.

```
class Department { }  
class College {  
    Department dept;  
}
```

Aggregation
has-a
(weak)

COUPLING VS COHESION



What is the difference between Coupling and Cohesion?

Ans: Coupling: Dependency between classes (less is better). Cohesion: Relatedness of functions within a class (more is better). High cohesion and low coupling make a good design.

Coupling

Low Coupling
= Good Design

Cohesion

High Cohesion
= Good Design

Remember:
High Cohesion
Low Coupling
= Best!

Understand Concepts → Practice Code → Crack Interviews! 😊

Practice Today
Crack Tomorrow!
😊

IMPORTANT DSA INTERVIEW QUESTIONS



★ **Arrays**

10 20 30 40

● **What are arrays? How are they different from linked lists?**

Discuss advantages, disadvantages and use cases of arrays.

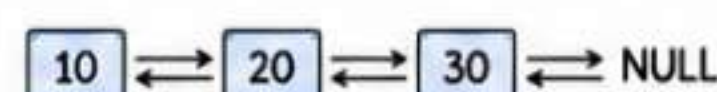


★ **Linked List**

10 → 20 → 30 → NULL

● **What is a linked list? What are its types?**

Explain singly, doubly, and circular linked lists with use cases.

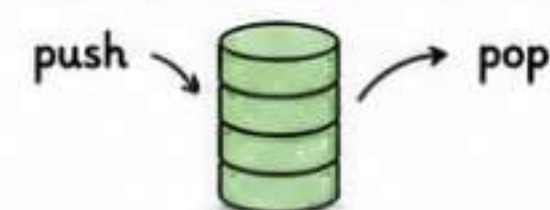


★ **Stacks**

Top

● **What is a stack? Explain its operations.**

How is stack used in recursion and expression evaluation?

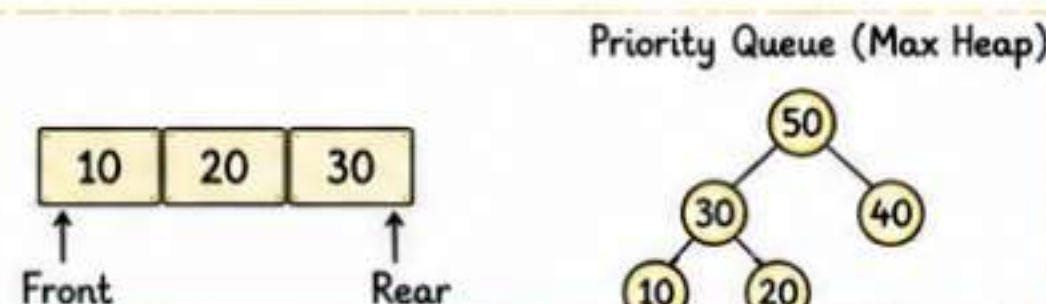


★ **Queues**

Front 10 20 30 Rear

● **What is a queue? What are its types?**

Explain linear queue, circular queue and priority queue with examples.

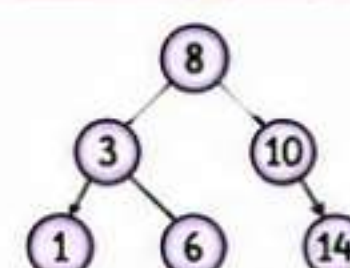


★ **Trees**

1
2 3
4 5 6

● **What is a tree? What are its types?**

Explain binary tree, binary search tree (BST) and balanced tree.

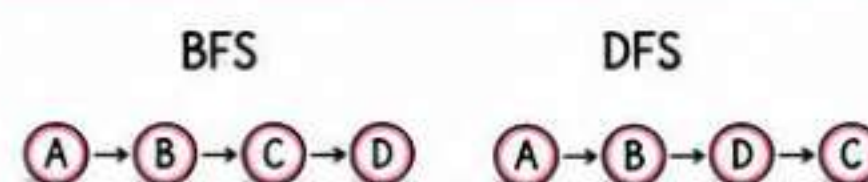


★ **Graphs**

A B C

● **What is a graph? How is it represented?**

Explain BFS and DFS with time and space complexities.



★ **Sorting & Searching**

● **What are the different sorting algorithms?**

Compare time complexities of Bubble Sort, Merge Sort and Quick Sort.

Bubble $O(n^2)$	Merge Sort $O(n \log n)$	Quick Sort $O(n \log n)$
--------------------	-----------------------------	-----------------------------

★ **Time & Space Complexity**

● **What is time complexity? What are its cases?**

Explain Big-O notation and time complexity cases with examples.

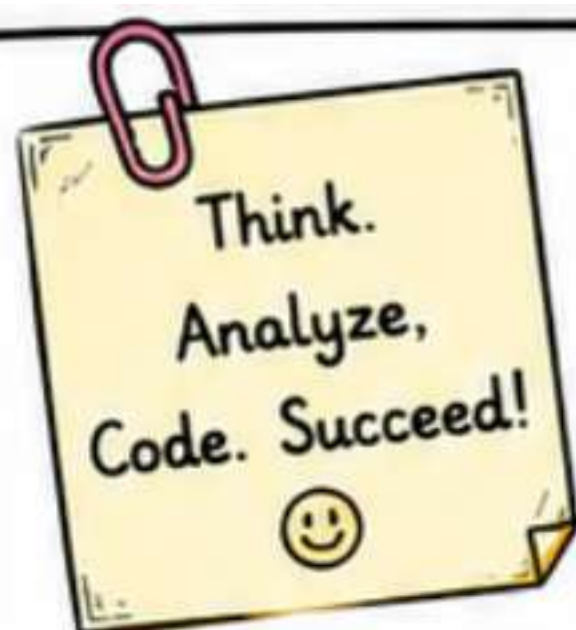
Common Complexities
 $O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$, $O(2^n)$, $O(n!)$

Topics Covered

- ✓ Arrays
- ✓ Linked List
- ✓ Stack
- ✓ Queue
- ✓ Trees
- ✓ Graphs
- ✓ Sorting
- ✓ Searching
- ✓ Time & Space Complexity

Keep Practicing
Keep Improving!
😊



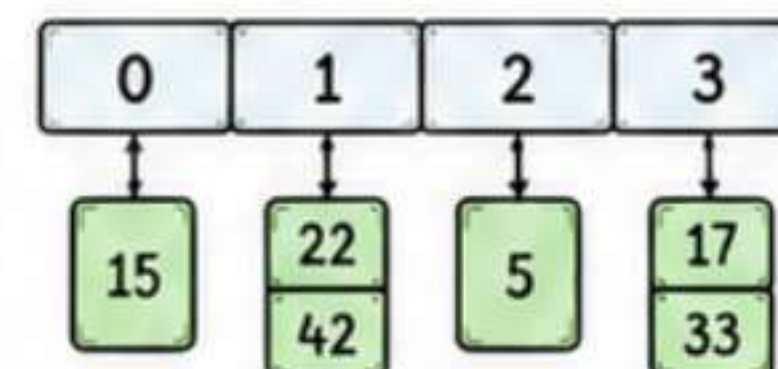


MORE IMPORTANT DSA INTERVIEW QUESTIONS



How does Hashing work? What are its applications?

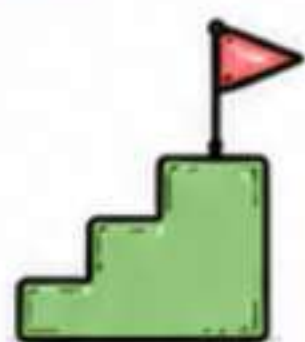
Explain hash functions, collision handling techniques and real-life applications.



What is Dynamic Programming?

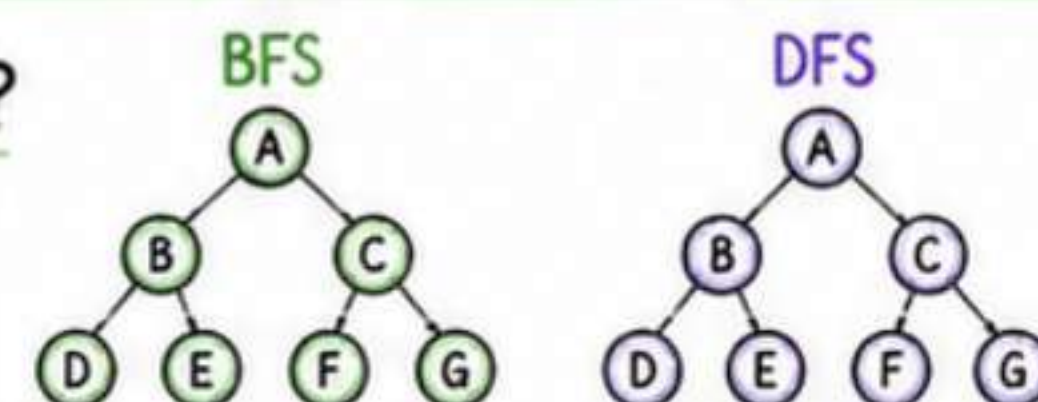
Explain with an example. How is it different from Greedy approach?

Overlapping Subproblems + Optimal Substructure



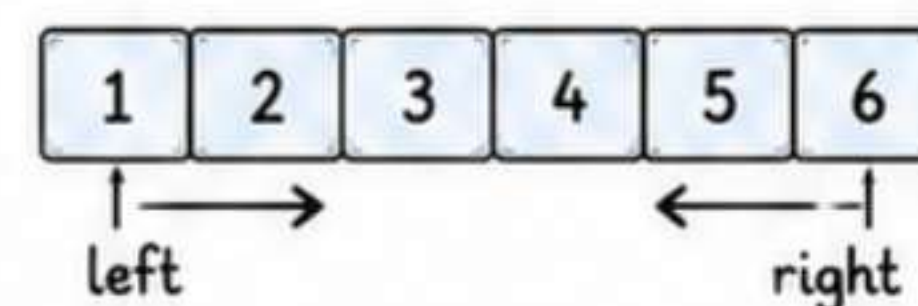
What is the difference between BFS and DFS?

Compare BFS and DFS in terms of approach, use cases, and space complexity.



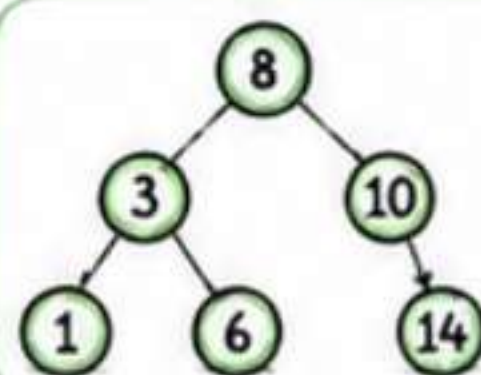
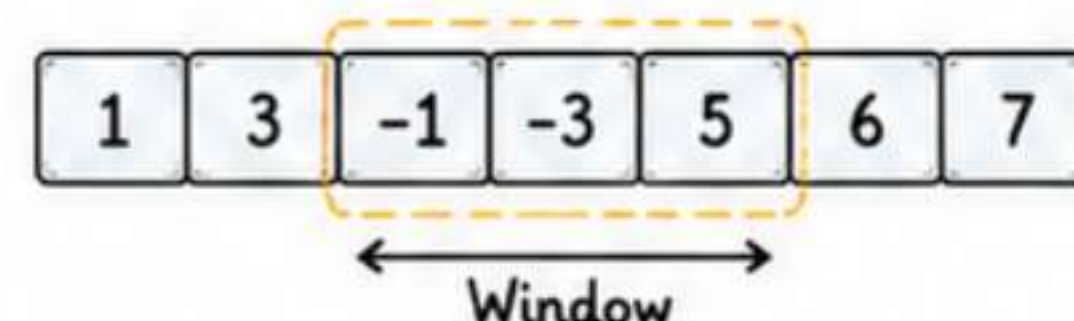
What is the Two Pointer Technique?

Explain the technique with examples and problems where it is used.



What is Sliding Window Technique?

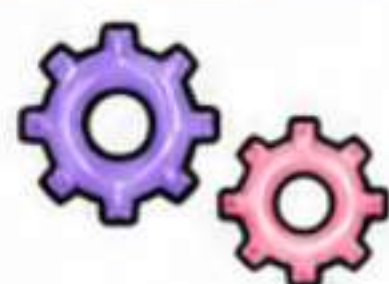
Explain the concept and give examples of problems.



What is a Binary Search Tree?

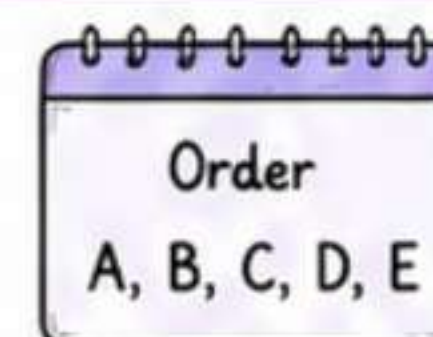
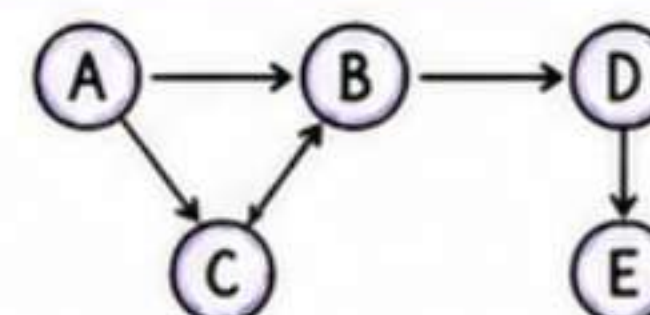
Explain its properties, operations and time complexity.

Properties:
✓ Left < Root < Right
✓ Inorder gives sorted order



What is Topological Sorting?

Where is it used? Explain with an example.



What is the difference between Array and Linked List?

Compare them based on memory, insertion, deletion and access.

Array	VS	Linked List
• Fixed size		• Dynamic size
• Contiguous memory		• Non-contiguous
• $O(1)$ access		• $O(n)$ access



How do you find the Kth largest/smallest element in an array?

Explain different approaches and their time complexities.

Use Sorting?
Or Heap?
Or QuickSelect?



Practice → Solve → Analyze → Improve → Ace the Interview!



Code Today
Crack Tomorrow!

MORE IMPORTANT DSA INTERVIEW QUESTIONS



String

Aa

What are common string algorithms?

Pattern searching (KMP, Rabin-Karp),
Longest Palindromic Substring,
String matching, Anagram check.



Pattern Search (KMP)

Text : A B A B A C A B A B A

Pat : A B A B A

Recursion
& Backtracking



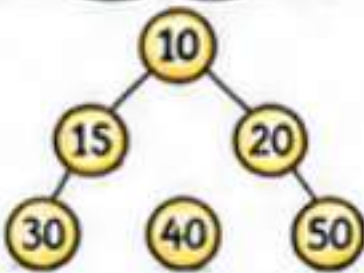
What is recursion? Give examples.

Explain recursion, base condition,
stack overflow.
Examples: Factorial, Fibonacci.

factorial(4)
= 4 * factorial(3)
= 4 * 3 * 2 * 1
= 24

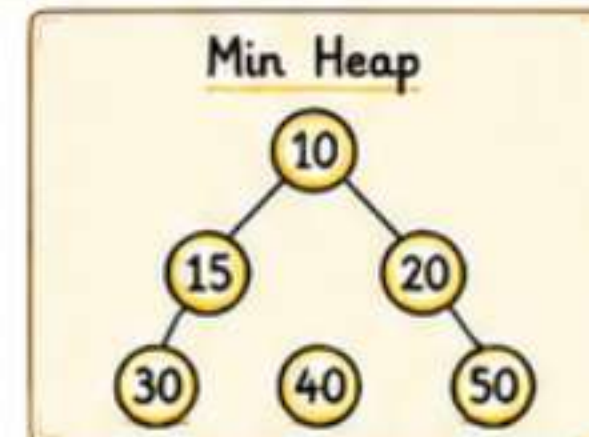
Backtracking
Examples:
N-Queens, Sudoku,
Subsets

Heap
(Priority Queue)



What is a heap? How is it used?

Types of heaps, heap operations (insert,
delete, heapify), and applications
like priority queue.



Heap Operations

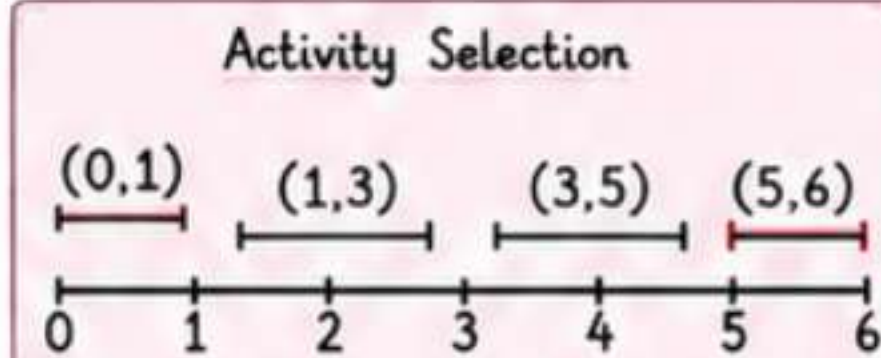
- ✓ Insert
- ✓ Delete
- ✓ Heapify

Greedy
Algorithm



What is Greedy Algorithm?

Explain greedy choice property.
Examples: Activity Selection,
Huffman Coding, Fractional Knapsack.



Greedy ≠
Always Optimal
But often
Effective!

DP
(Dynamic Prog.)

Time	Mem	Code
...	...	...

What is Dynamic Programming?

Explain overlapping subproblems
and optimal substructure.
Examples: 0/1 Knapsack, LCS, Edit Distance.

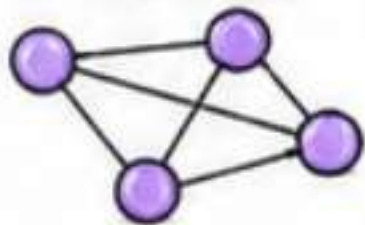
Fibonacci (Top-Down DP)

fib(5) = fib(4) + fib(3)
fib(4) = fib(3) + fib(2)
...
fib(1) = 1, fib(0) = 0

Memo Table

n	fib(n)
0	0
1	1
2	1
3	2
4	3
5	5

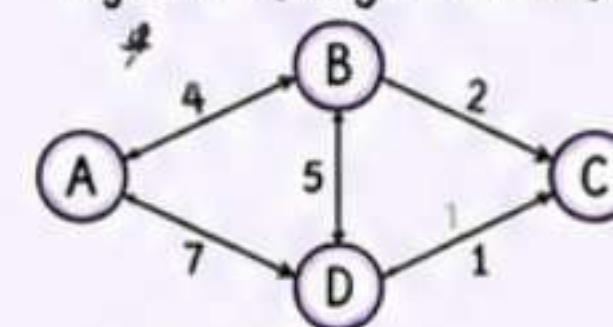
Graph
Algorithms



What are shortest path algorithms?

Explain Dijkstra's Algorithm
and Bellman-Ford Algorithm with
use cases.

Dijkstra (Single Source)



Use Cases

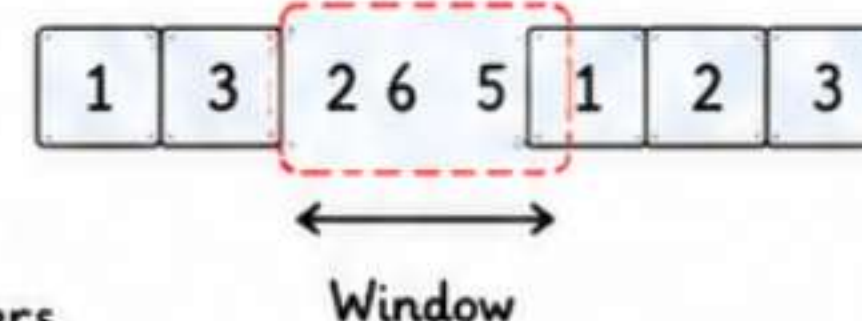
- GPS Navigation
- Network Routing
- Social Networks

Sliding Window
Technique



What is sliding window technique?

Fixed size and variable size window.
Examples: Max Sum Subarray,
Longest Substring Without Repeating Characters.



Efficient for
array/string
problems!

Bit
Manipulation



What are bit manipulation basics?

AND, OR, XOR, LEFT/RIGHT SHIFT,
Check power of 2, Set/Clear bits.

Examples

- $n \& (n-1) \rightarrow$ Clear last set bit
- $n \& (n-1) == 0 \rightarrow$ Power of 2
- $n \wedge n = 0, n \wedge 0 = n$

Useful in
Optimization
& Competitive
Programming!



Practice Regularly $\rightarrow$ Solve Problems $\rightarrow$ Analyze $\rightarrow$ Improve $\rightarrow$ Ace the Interview! 😊



Consistency
today,
Success
tomorrow!



CONCLUSION

INTERVIEW QUESTIONS

OOPS • DSA • JAVA



KEY TAKEAWAYS

- ✓ OOPS helps in writing clean, modular and reusable code.
- ✓ DSA helps in solving problems efficiently and optimally.
- ✓ Java concepts form the foundation for building robust applications.
- ✓ Practice concept + code + problem solving regularly.
- ✓ Understand deeply, don't just memorize.
- ✓ Explain with examples, draw diagrams and write clean code.



OOPS

- Focus on concepts like Inheritance, Polymorphism, Abstraction, Encapsulation.
- Understand relationships between classes.
- Know access modifiers, constructors, this & super, static, final.
- Be ready with real-time examples.
- Think in terms of objects and interactions.



DSA



- Strengthen fundamentals: Arrays, Strings, Linked List, Stacks, Queues.
- Practice Trees, Graphs, Heaps, Hashing.
- Master algorithms: Sorting, Searching, Greedy, DP, Backtracking.
- Analyze time & space complexity.
- Solve problems on paper first, then code.



JAVA



- Strong on OOPs concepts in Java.
- Know core Java: Collections, Exception Handling, Multithreading, I/O.
- Understand JVM, Memory Management, Garbage Collection.
- Practice coding and write clean, efficient code.
- Stay updated with latest Java features.



FINAL TIPS

- ♥ Revise concepts regularly.
- ♥ Solve previous year questions.
- ♥ Mock interviews build confidence.
- ♥ Communicate your thought process clearly.
- ♥ Stay calm, think before you code.
- ♥ Keep learning and keep improving!



It's not about knowing everything,
it's about learning, practicing and improving every day.

YOU'VE GOT THIS! ♥

